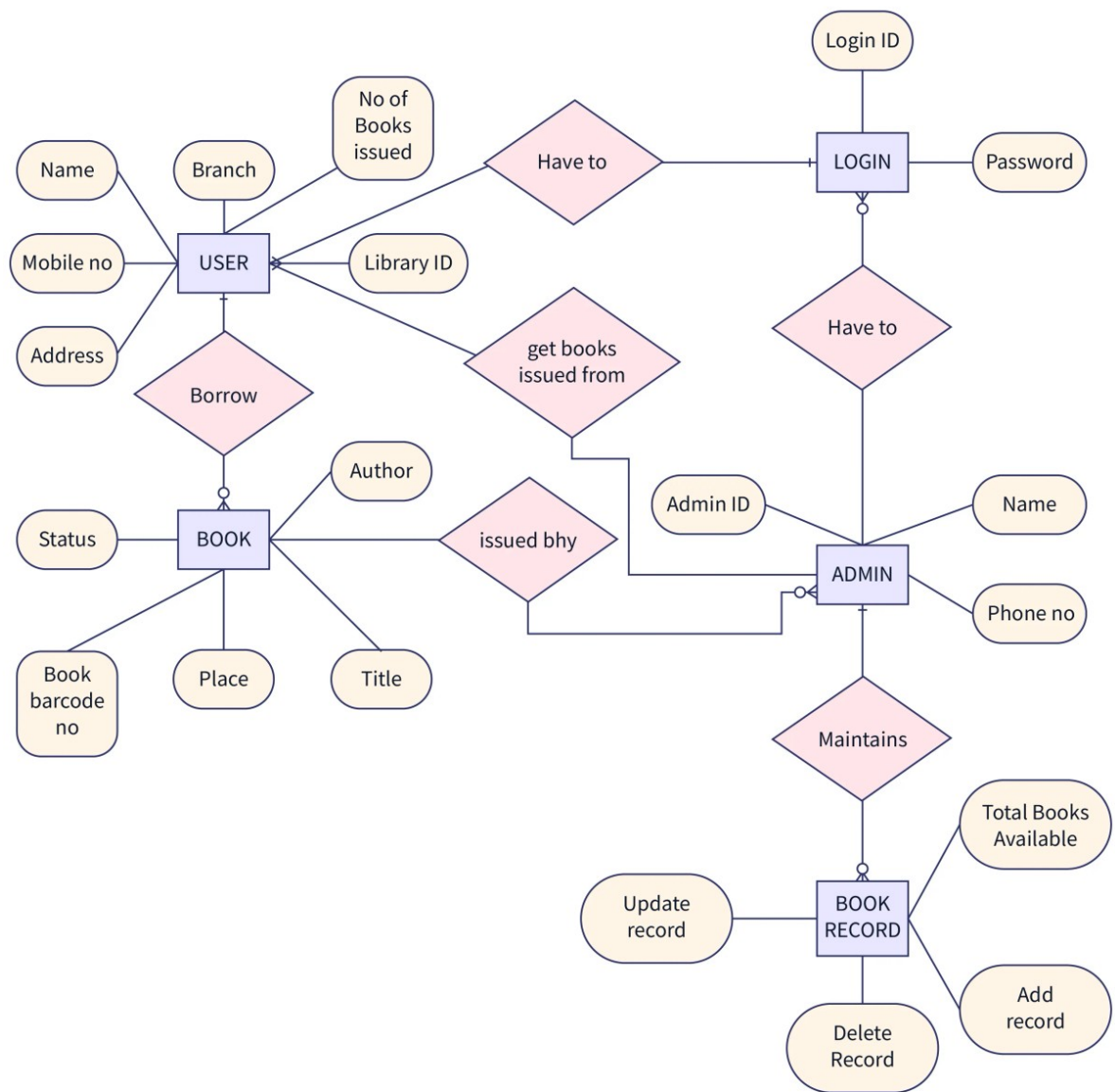
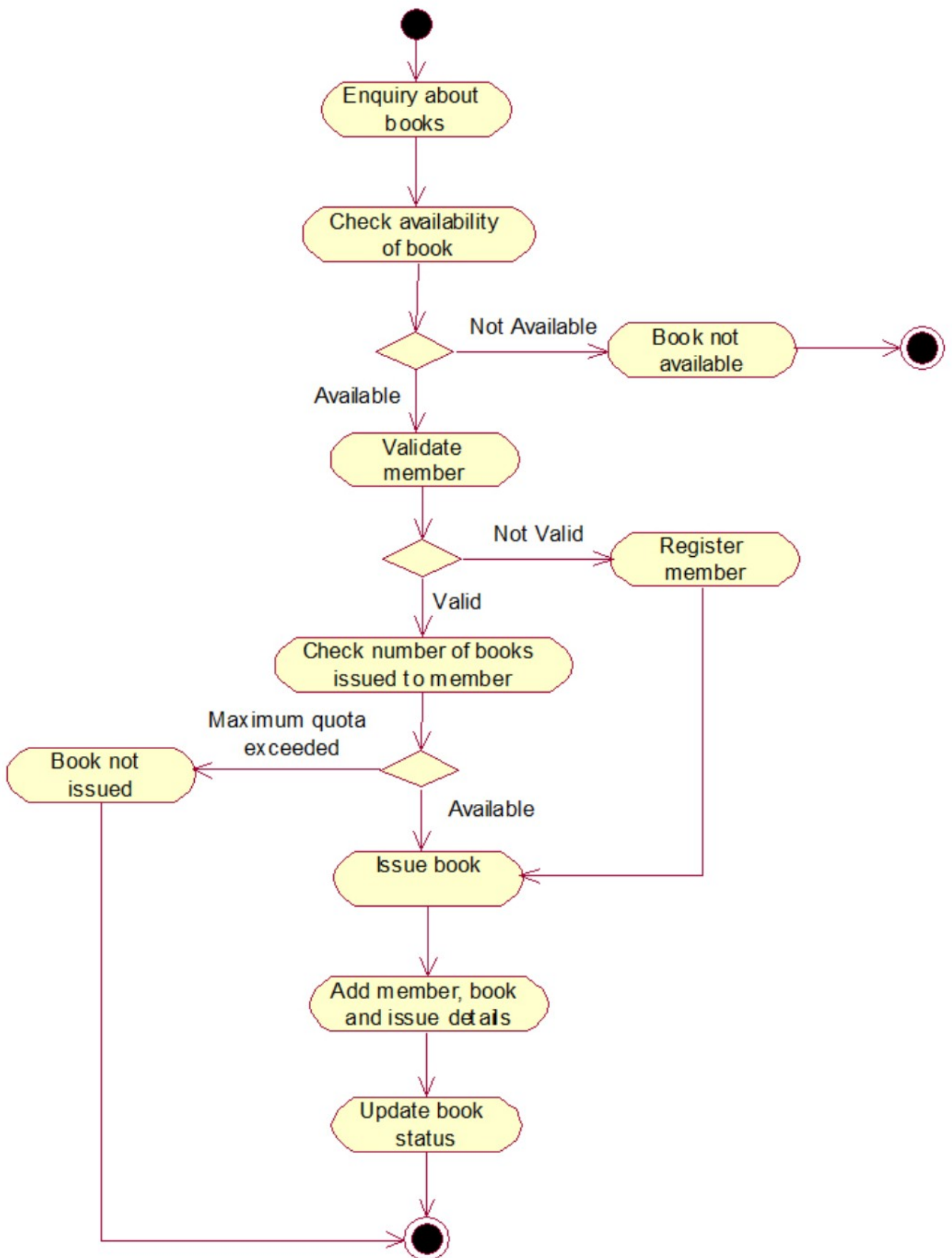






# INTEGRATED LIBRARY MANAGEMENT SYSTEM ALGORITHM

VERSION 3 • FLOWCHART — SYSTEM PROCESS DESIGN

## PHASE 1 DATABASE SETUP

### Initialize Database

#### Create 5 Tables

- **students**     student\_id, name, contact, status
- **books**        book\_id, title, author, available\_count
- **loans**         loan\_id, student\_id, book\_id, issue\_date, due\_date
- **reservations**    res\_id, student\_id, book\_id, status
- **finest**         fine\_id, loan\_id, amount, paid\_status

## PHASE 2 CRUD: STUDENT & BOOK MODULES

**Student Module CRUD**

- Create / Read / Update / Delete Student Records

**Book Module CRUD**

- Create / Read / Update / Delete Book Records

## PHASE 3 CIRCULATION

### Start Circulation Process

**Check Availability**  
Is book.available\_count > 0 ?

YES

NO

### Issue Book

Decrement available\_count by 1

### Create Loan Record

- Record loan details & set due\_date

**Issue Successful** ✓ End

**Reservation Active**  
Add reservation for student

### RETURN PROCESS

### Return Book

**Fine = Days Overdue × Rate**  
Fine = (Days Overdue) × (Fine Rate per Day)  
if Days Overdue > 0, else Fine = 0

- Update Loan as Returned** ↑
- Mark loan as returned, update return\_date

**Return Completed** ✓ End